

Requirements Table :

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow donors to post surplus food packages with perishable expiry countdown timers and dietary tags.	High	Pass: A donor can successfully post a food batch with expiry time and dietary tags. Fail: The batch cannot be posted when required details are missing.	Core functionality for making surplus food available to shelters.
FR-002	Functional	The system shall allow verified shelter coordinators to view available surplus food batches before their expiry time.	High	Pass: A verified shelter coordinator can view active, non-expired food batches. Fail: An expired batch appears as available.	Allows shelters to identify food that can still be collected.
FR-003	Functional	The system shall allow verified shelter coordinators to reserve an available food batch and select a pickup window.	High	Pass: A shelter can reserve an available batch and receive a confirmed pickup window. Fail: The system allows two shelters to reserve the same batch.	Ensures surplus food is allocated to a specific shelter.
FR-004	Functional	The system shall generate a pickup verification code for a confirmed food reservation.	High	Pass: A unique verification code is generated after successful reservation. Fail: No verification code is generated for a confirmed reservation.	Provides a simple way to verify the authorized pickup.
FR-005	Functional	The system shall automatically deactivate a food batch when it expires or after it has been successfully collected.	High	Pass: An expired or collected batch is removed from the active claim board. Fail: An expired or collected batch remains available for reservation.	Prevents expired or unavailable food from being claimed.

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NFR-001	Non-Functional	The platform shall dispatch instant push notifications to shelters within a 5 km radius when high-quantity perishable food is listed.	High	Pass: Benchmarking confirms notifications reach eligible shelters within the required target latency under simulated peak load. Fail: Notifications exceed the target latency or are sent outside the defined 5 km radius.	Helps nearby shelters respond quickly to highly perishable food.
NFR-002	Non-Functional	The platform shall restrict food reservation and pickup information to verified and authorized shelter coordinators.	High	Pass: Only verified shelter coordinators can access reservation and pickup information. Fail: An unverified user can access or reserve food.	Protects reservation information and prevents unauthorized food claims.